

MOHAMED RAGAB

AI Engineer | Construction & Infrastructure

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Military status: Exempted

PROFESSIONAL SUMMARY

AI engineer with a structural engineering background, specialising in artificial intelligence for construction and infrastructure. I develop computer vision and machine learning solutions that detect structural defects in site photos, read structural drawings and convert laser scans into BIM models, taking each one from research and data preparation through model training to cloud deployment. Experienced in BIM delivery to ISO 19650 on UK projects and in structural design for US projects. MSc in Digital Construction Analytics & BIM with Distinction from Ulster University, UK.

CORE COMPETENCIES

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|-------------------------|-----------------------------------|-------------------------------|
| • AI & Machine Learning | • Computer Vision & Deep Learning | • Model Training & Evaluation |
| • Data Engineering | • Cloud AI (RunPod, Azure) | • AI-Assisted Development |
| • BIM & Digital Twins | • Scan-to-BIM & CAD-to-BIM | • Structural Engineering |

PROFESSIONAL EXPERIENCE

R&D AI-Construction Specialist | AGECS Apr 2026 – Present
Remote, part-time

- Develop AI models that automatically detect structural elements (columns, beams, walls, openings and piles) on structural drawings, as the foundation for converting 2D drawings into BIM models (CAD-to-BIM). The column model reaches over 93% precision and recall.
- Prepare and manage large training datasets: combining multiple public sources, cleaning and converting annotations, and quality-checking labels.
- Built a Scan-to-BIM pipeline that converts site laser scans into verified IFC models of slabs, walls and columns, automatically excluding temporary works.
- Evaluate the latest AI models, including multimodal AI, for understanding engineering drawings.

Chief Technology Officer (CTO) | AECAI Ltd Dec 2025 – Present
Belfast, UK · AI-assisted structural inspection platform

- Lead the company's technology and AI strategy, from research and model selection to production deployment.
- Train and deploy deep learning models that detect cracks, spalling and exposed reinforcement in inspection photos, running on cloud GPUs (RunPod).
- Built the web platform where inspectors upload photos, the AI flags defects, and engineers review the findings and issue reports.
- Established a rigorous data and evaluation process: dataset auditing, checks against data leakage, and cost-controlled GPU training. Improved post-processing cut false detections by two-thirds on the hardest test photo.
- Use AI coding tools (Claude Code, OpenAI Codex) to develop and ship software quickly, with code review and automated testing.

Digital Design & BIM Engineer | AECOM × Ulster University Feb 2025 – May 2025
Belfast, UK · industry project

- Led a digital-twin project for a concrete water tank: built Revit models from AECOM drawings and linked them to inspection data using Dynamo and IFC.
- Developed a Power BI dashboard showing the 3D model, a crack register and recommended maintenance actions based on Eurocode 2 limits.
- Trained a deep learning crack classifier to support remote inspection. Module grade: 80%.

- Modelled a multi-storey office building in Revit from PDF drawings and coordinated the structural, architectural and MEP models.
- Applied ISO 19650 information management: naming conventions, a common data environment (ACC / BIM 360) and the BIM Execution Plan.
- Ran clash detection in Navisworks and automated modelling tasks with Dynamo and pyRevit.

- Delivered more than 115 structural and façade design packages, including curtain walls, glass railings, canopies, stairs and connections.
- Designed in steel, aluminium and timber to AISC, ASCE 7, ACI and the Florida Building Code using ETABS, SAP2000, SAFE and IDEA StatiCa.

SELECTED PROJECTS

- Concrete Defect Detection with Deep Learning** (MSc research): compared leading deep learning models and validated them on real site photos from an industry partner; the best model found 93% of real defects in field tests. Journal paper under review.
- IFC Model Auditor** (open source): automated quality checking of BIM models against ISO 19650 and buildingSMART standards.
- Structural Engineering Tools** (open source): Eurocode 2 crack-width checks and column load calculations taken directly from BIM models.
- Construction Material Price Forecasting**: machine learning forecasts of UK material prices to support cost planning.

EDUCATION

TECHNICAL SKILLS

AI & Machine Learning	Python, PyTorch, TensorFlow / Keras, YOLO, U-Net, OpenCV, scikit-learn, Open3D
AI Platforms & Cloud	RunPod (GPU), Microsoft Azure, Hugging Face, Docker, Supabase; Claude, OpenAI and Gemini models
Software Development	Claude Code, OpenAI Codex, VS Code, Git / GitHub, GitHub Actions, TypeScript / Next.js, SQL
BIM & Digital Twins	Revit, Navisworks, AutoCAD, Dynamo, pyRevit, ACC / BIM 360, IFC, Power BI, Speckle
Structural Engineering	ETABS, SAP2000, SAFE, IDEA StatiCa, Tekla; Eurocodes, AISC, ASCE 7, ACI

CERTIFICATIONS & LANGUAGES

- Deep Learning for Computer Vision – ITI Mahara-Tech (2026) · AI Fluency – Anthropic (2026)
- Steel Design Diploma and Concrete Technical Design – Autodesk ATC (2022) · Tekla Structures (2024)
- Institution of Civil Engineers (ICE) – Student Member
- Languages: Arabic (native), English (fluent)